

JOHN ROHAL

MECHANICAL ENGINEERING | OHIO UNIVERSITY

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SKILLS

CAD

- SolidWorks (FEA, Sheet Metal, Drawings)
- AutoCAD, OnShape, Revit
- Mastercam, G90/G91

DESIGN & ANALYSIS

- DFM/DFA, DFMEA
- Stress & Fatigue Analysis
- Root Cause Analysis
- Rapid Prototyping (3D Printing)

MANUFACTURING

- Dimensional Tolerancing
- CNC Milling, Turning, Sinker EDM
- Soldering, Wiring, Circuit Theory

SOFTWARE

- MATLAB, Python, EES
- HTML, CSS, JavaScript

KEY PROJECTS

Freight Elevator | Jan – May 2026

- Redesigned the drivetrain, support brackets, and loading platform to improve load capacity and efficiency.
- Validated redesigns using FEA and hand calculations with a maximum percent difference of 1.06%.

Carrier Shafts | Nov – Dec 2025

- Designed shafts using bending stress analysis, S-N fatigue methods, and Modified Goodman criteria.
- The design supported a 3-ton gantry trolley with infinite life in fatigue and a FOS of 3.5.

Battle Bot | Jan – May 2024

- Advanced to competition finals and received “Most Creative Design” award.
- Designed the chassis and weapon systems in SolidWorks.
- Fabricated all components using additive manufacturing before assembly and electrical wiring.

EDUCATION

OHIO UNIVERSITY

BS in Mechanical Engineering

Aug 2023 – May 2027

- 3.63/4.00 GPA
- Mathematics Minor
- Lean-Six Sigma Certification

EXPERIENCE

Ceco Concrete Construction | Cincinnati, OH | May – Aug 2026

Project Manager – Intern

- Facilitated preconstruction planning for the \$2.5B Cleveland Browns stadium project through quantity takeoffs, cost estimating, and schedule planning using PlanSwift, Bluebeam Revu, and Spitfire.
- Supported FEA-driven structural analysis, Revit-based formwork design, and bill of materials management while coordinating with engineering and field teams.
- Conducted jobsite inspections of rigging, tools, fall protection systems, and safety equipment to support OSHA-compliant operations.

Amazon | Imperial, PA | May – Aug 2025

Equipment Operator – Full-Time

- Operated powered industrial trucks (PITs) to move, stage, and organize 300+ inventory units per shift while maintaining a 100% accident-free safety record.
- Cross-trained across multiple warehouse operations to support inventory staging and logistics within a high-volume fulfillment center.

ODOT | New Philadelphia, OH | May – Aug 2024

Construction Engineer – Intern

- Led and supervised a materials coring and sampling crew, ensuring efficient and safe operations.
- Performed field testing of asphalt and concrete while documenting results to verify specification compliance.
- Managed contract line-item summaries and change orders using AWP software, ensuring accurate project documentation and cost tracking.

Amazon | Imperial, PA | May – Aug 2023

Fulfillment Center Warehouse Associate – Part-Time

- Contributed to a 3% increase in the processing rate of the facility, highlighting commitment to productivity and process improvement.
- Consistently recognized as one of the facility’s top processors.

EXTRACURRICULARS

SAE Baja | Athens, OH | Jan – May 2025

Manufacturing Team Member

- Modeled mechanical components in OnShape while collaborating with manufacturing team members during vehicle development.
- Gained hands-on experience with design workflows and iterative product development in a competitive engineering environment.

Ohio University Marching 110 | Athens, OH | Aug 2024 – Present

Member & Staff

- Performed as a member of a nationally recognized collegiate marching band, demonstrating leadership, teamwork, discipline, and reliability.
- Silver Coin Recipient, awarded to those who have helped the band achieve greater heights through performance, accomplishment, support, and service.